Reproducibility Report for Study by Salamons et al. (2026, Animal Behaviour)

Reproduce Figure 2

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Load

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to beco
```

```
library(readxl)
install.packages("patchwork")
```

```
Error in `contrib.url()`:
! trying to use CRAN without setting a mirror
```

```
  library(patchwork)
install.packages("lme4")
```

```
Error in `contrib.url()`:
! trying to use CRAN without setting a mirror
```

```
  library(lme4)
```

```
Loading required package: Matrix

Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

    expand, pack, unpack
```

```
auditory <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Auditory_Discr
```

```
Rows: 497 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
causal <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Causal_Reasoning
```

```
Rows: 500 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
```

```
chr (4): Dog Name, Status, Rearing, Sex
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
detour <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Cylinder_Inhibit
```

```
Rows: 498 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
reversal <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Cylinder_Revers
```

```
Rows: 496 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
marker <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Marker_Gesture.cs
```

```
Rows: 503 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
momentary <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Momentary_Poi
```

```
Rows: 495 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
odor <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Odor_Discriminatio
```

```
Rows: 492 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
unsolvable <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Unsolvable_Ey
```

```
Rows: 498 Columns: 12
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (8): Dog ID, Age (wks), Age (days), Total Trials, Mean Look Time, # Tria...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
workingmem <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Working_Memo
```

```
Rows: 493 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
```

```
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
distraction <- read_csv ("~/Documents/salomons2026/Data/CleanedTests/CleanedData_Working_Memo
```

```
Rows: 502 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): Dog Name, Status, Rearing, Sex
dbl (7): Dog ID, Age (wks), Age (days), Total Trials, Trials Correct, Propor...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Individual Plots

```
auditory_long <- auditory |>
  filter(Status=="Longitudinal") |>
   mutate(
    `Age (wks)` = as.numeric(`Age (wks)`),
    `Proportion Correct` = as.numeric(`Proportion Correct`),
     `experience` = as.numeric(`# Prior Test Sessions`)
  )

auditory_plot <- auditory_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25),
    labels = as.character
```

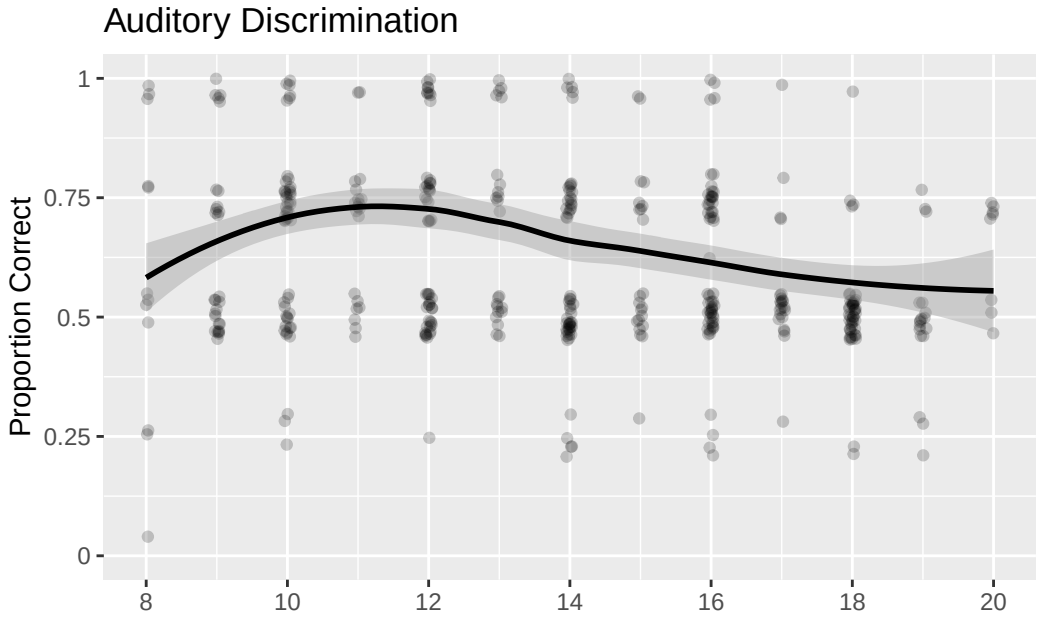
```
  ) +
  labs(title = "Auditory Discrimination")

print(auditory_plot)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 70 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
causal_long <- causal |>
  filter(Status=="Longitudinal") |>
   mutate(
    `Age (wks)` = as.numeric(`Age (wks)`),
    `Proportion Correct` = as.numeric(`Proportion Correct`),
     `experience` = as.numeric(`# Prior Test Sessions`)
  )
```

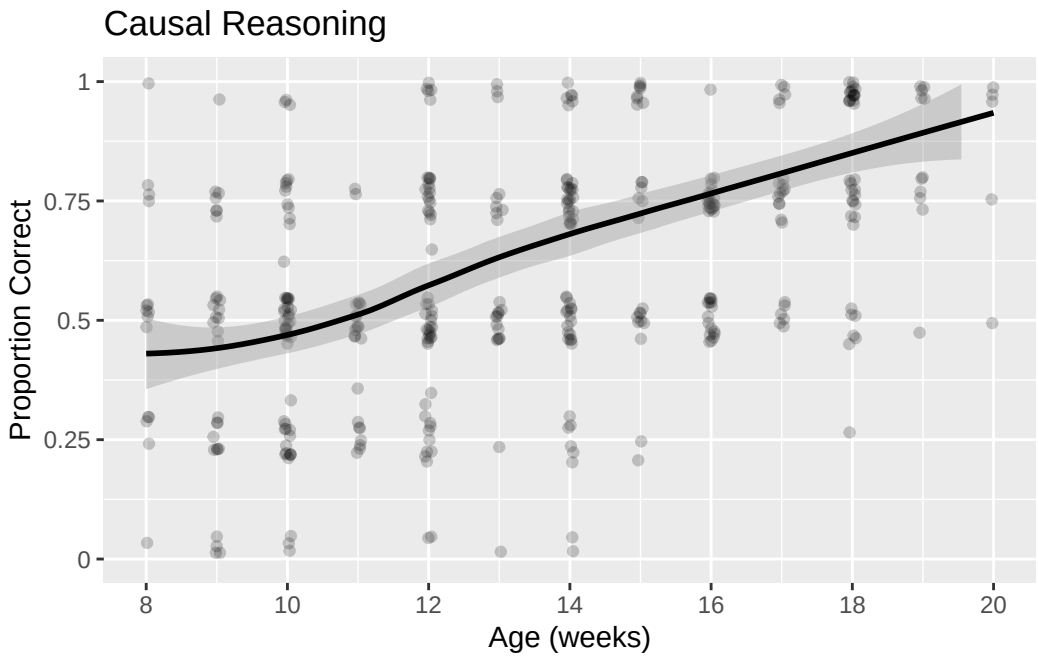
```r
causal_plot <- causal_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "Age (weeks)",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25),
    labels = as.character
  ) +
  labs(title = "Causal Reasoning")

print(causal_plot)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 98 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
detour_long <- detour |>
  filter(Status=="Longitudinal") |>
   mutate(
    `Age (wks)` = as.numeric(`Age (wks)`),
    `Proportion Correct` = as.numeric(`Proportion Correct`),
     `experience` = as.numeric(`# Prior Test Sessions`)
  )

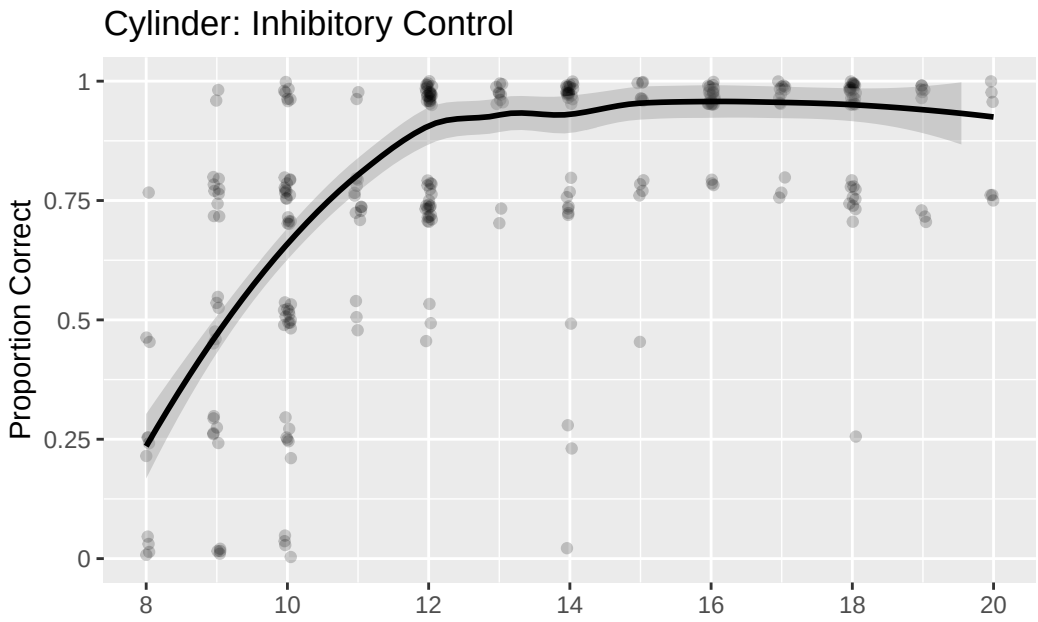
detour_plot <- detour_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25),
    labels = as.character
  ) +
  labs(title = "Cylinder: Inhibitory Control")
```

```
print(detour_plot)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 195 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
distraction_long <- distraction |>
  filter(Status=="Longitudinal") |>
   mutate(
    `Age (wks)` = as.numeric(`Age (wks)`),
    `Proportion Correct` = as.numeric(`Proportion Correct`),
     `experience` = as.numeric(`# Prior Test Sessions`)
  )

distraction_plot <- distraction_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
```

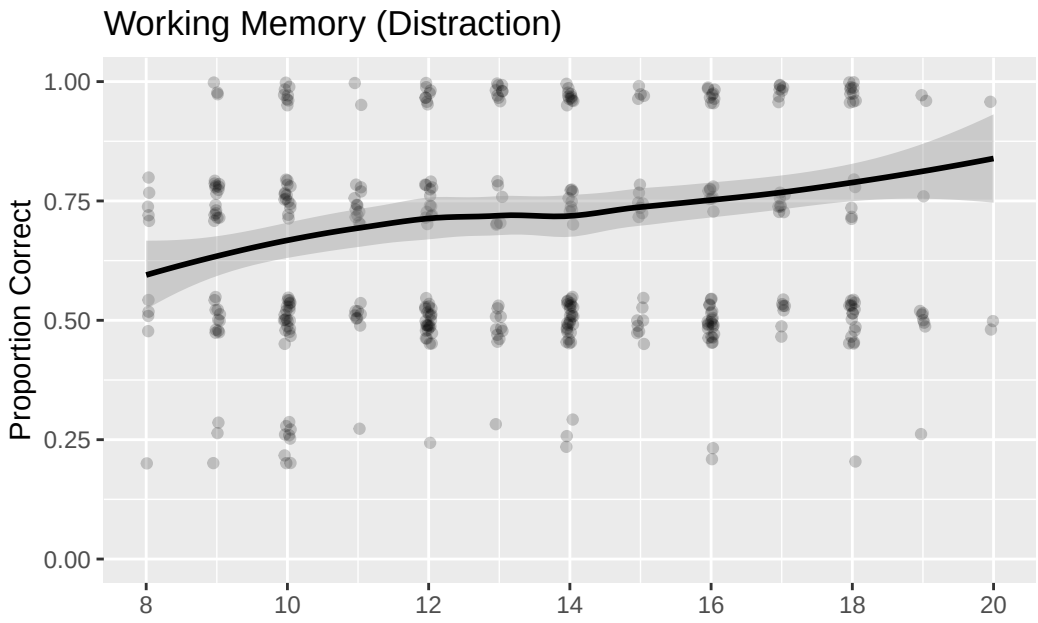
```
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2),
    labels = as.character
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25)
  ) +
  labs(title = "Working Memory (Distraction)")

print(distraction_plot)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 113 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
marker_long <- marker |>
  filter(Status=="Longitudinal") |>
   mutate(
     `Age (wks)` = as.numeric(`Age (wks)`),
     `Proportion Correct` = as.numeric(`Proportion Correct`),
      `experience` = as.numeric(`# Prior Test Sessions`)
  )

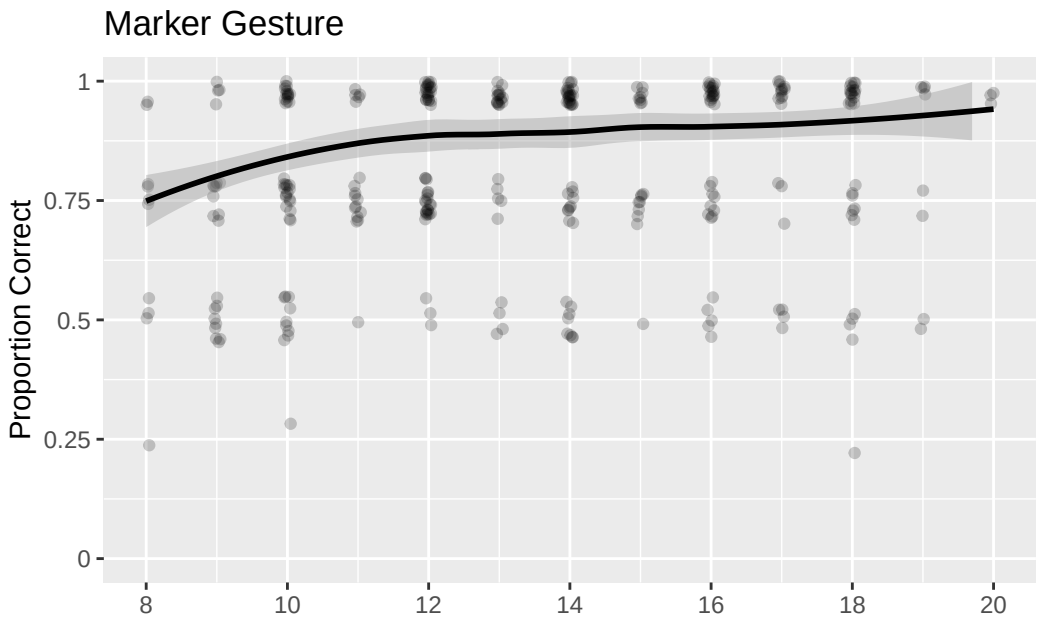
marker_plot <- marker_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25),
    labels = as.character
  ) +
  labs(title = "Marker Gesture")
```

```
print(marker_plot)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 172 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
momentary_long <- momentary |>
  filter(Status=="Longitudinal") |>
   mutate(
    `Age (wks)` = as.numeric(`Age (wks)`),
    `Proportion Correct` = as.numeric(`Proportion Correct`),
     `experience` = as.numeric(`# Prior Test Sessions`)
  )

momentary_plot <- momentary_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
```

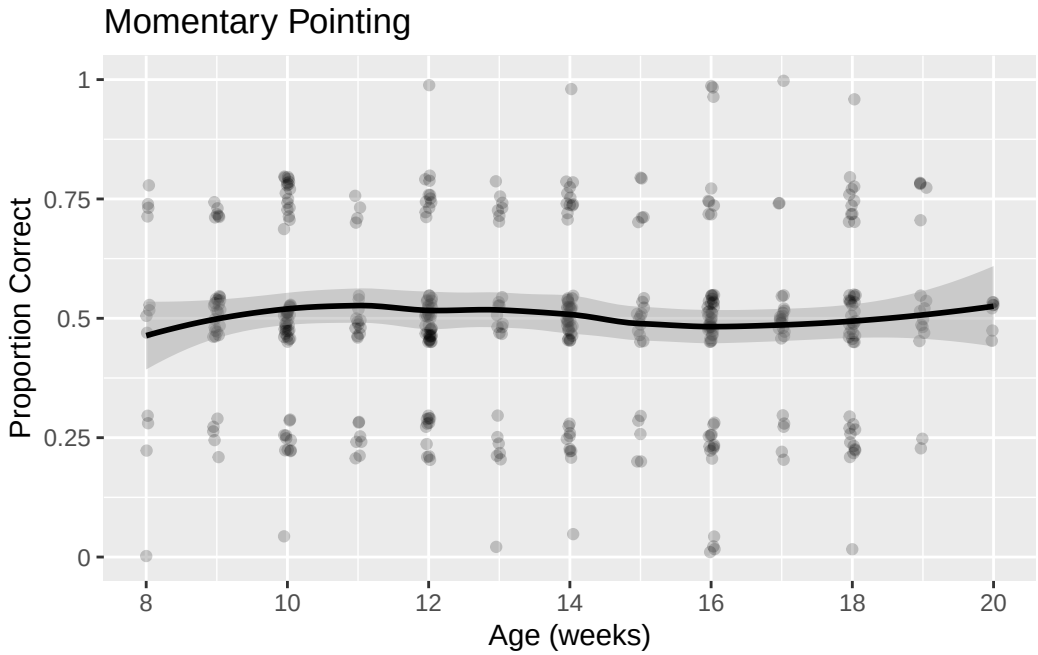
```
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "Age (weeks)",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25),
    labels = as.character
  ) +
  labs(title = "Momentary Pointing")

print(momentary_plot)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 38 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
odor_long <- odor |>
  filter(Status=="Longitudinal") |>
   mutate(
     `Age (wks)` = as.numeric(`Age (wks)`),
     `Proportion Correct` = as.numeric(`Proportion Correct`),
      `experience` = as.numeric(`# Prior Test Sessions`)
  )

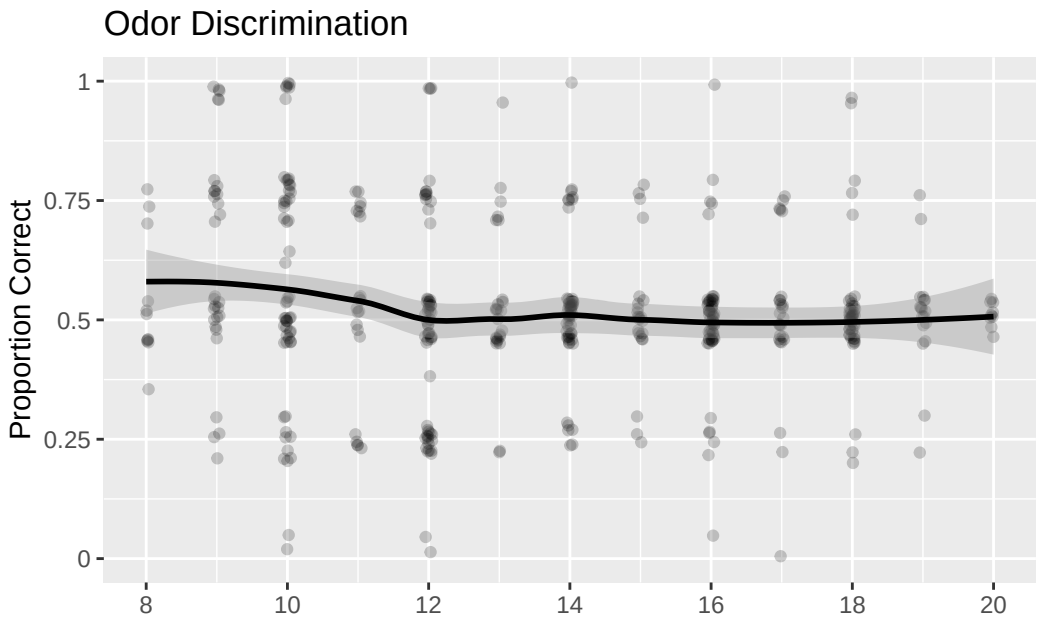
odor_plot <- odor_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25),
    labels = as.character
  ) +
  labs(title = "Odor Discrimination")
```

```
print(odor_plot)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 35 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
reversal_long <- reversal |>
  filter(Status=="Longitudinal") |>
   mutate(
    `Age (wks)` = as.numeric(`Age (wks)`),
    `Proportion Correct` = as.numeric(`Proportion Correct`),
     `experience` = as.numeric(`# Prior Test Sessions`)
  )

reversal_plot <- reversal_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
```

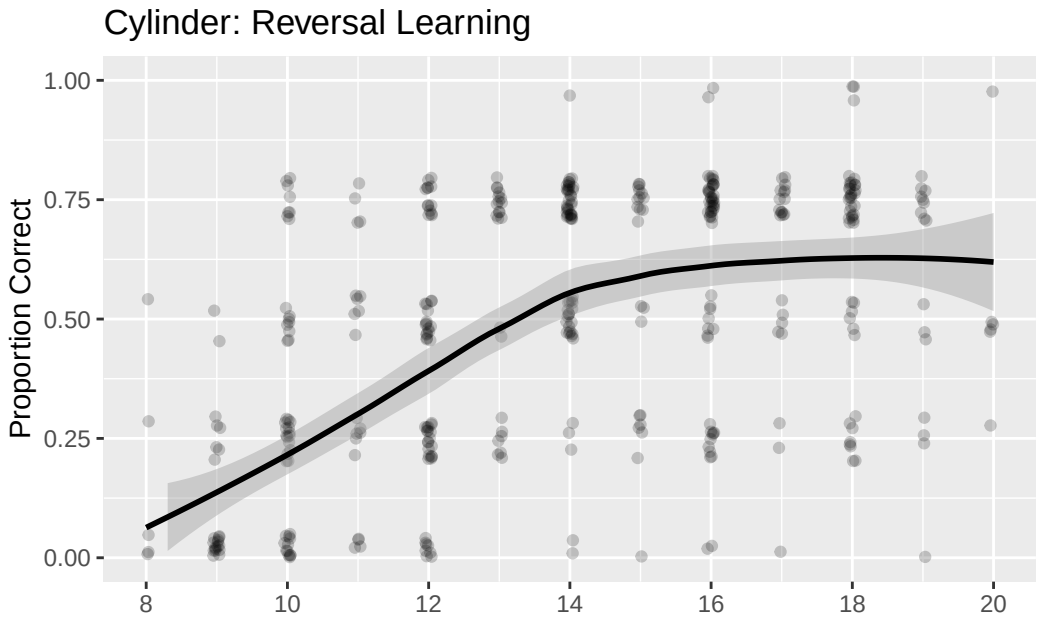
```
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25)
  ) +
  labs(title = "Cylinder: Reversal Learning")

print(reversal_plot)
```

```
`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 84 rows containing missing values or values outside the scale range
(`geom_point()`).
```



```r
unsolvable_long <- unsolvable |>
  filter(Status=="Longitudinal") |>
   mutate(
    `Age (wks)` = as.numeric(`Age (wks)`),
    `Proportion Trials Looked` = as.numeric(`Proportion Trials Looked`),
    `Mean Look Time` = as.numeric(`Mean Look Time`),
     `experience` = as.numeric(`# Prior Test Sessions`)
  )

proplooked_plot <- unsolvable_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Trials Looked`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "Age (weeks)",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25),
    labels = as.character
  ) +
  labs(title = "Unsolvable (Proportion Looked)")

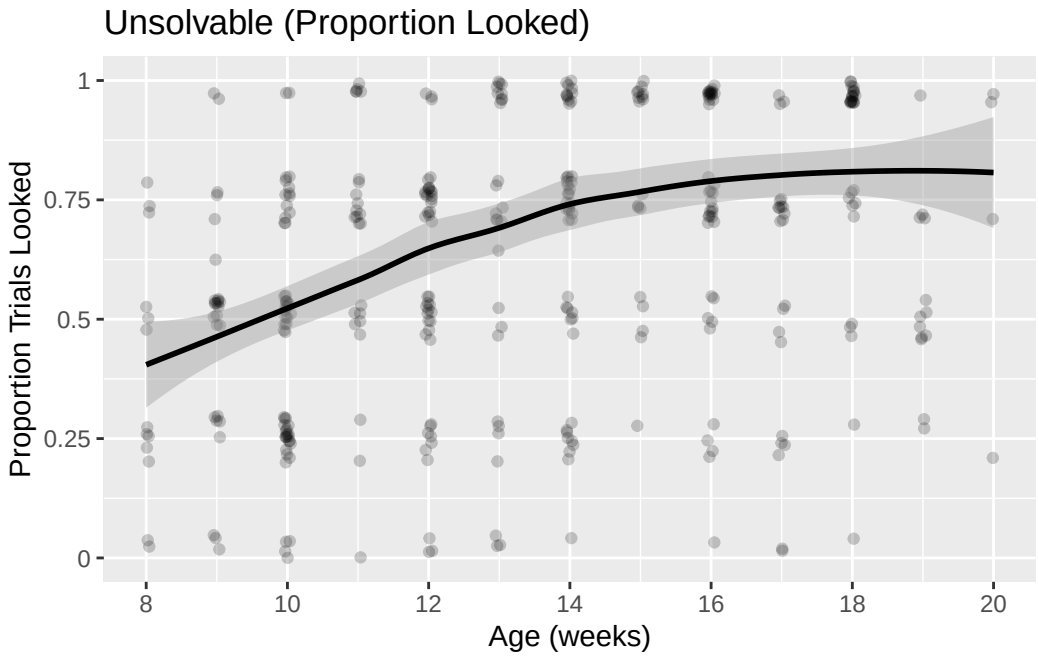
looktime_plot <- unsolvable_long|>
  ggplot(aes(x=`Age (wks)`, y = `Mean Look Time`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name = "",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    name = "Mean Look Time (s)",
    limits = c(0, 20),
    breaks = seq(0, 20, by = 5)
  ) +
  labs(title = "Unsolvable (Looking Time)")

print(proplooked_plot)
```

```
`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 117 rows containing missing values or values outside the scale range
(`geom_point()`).
```

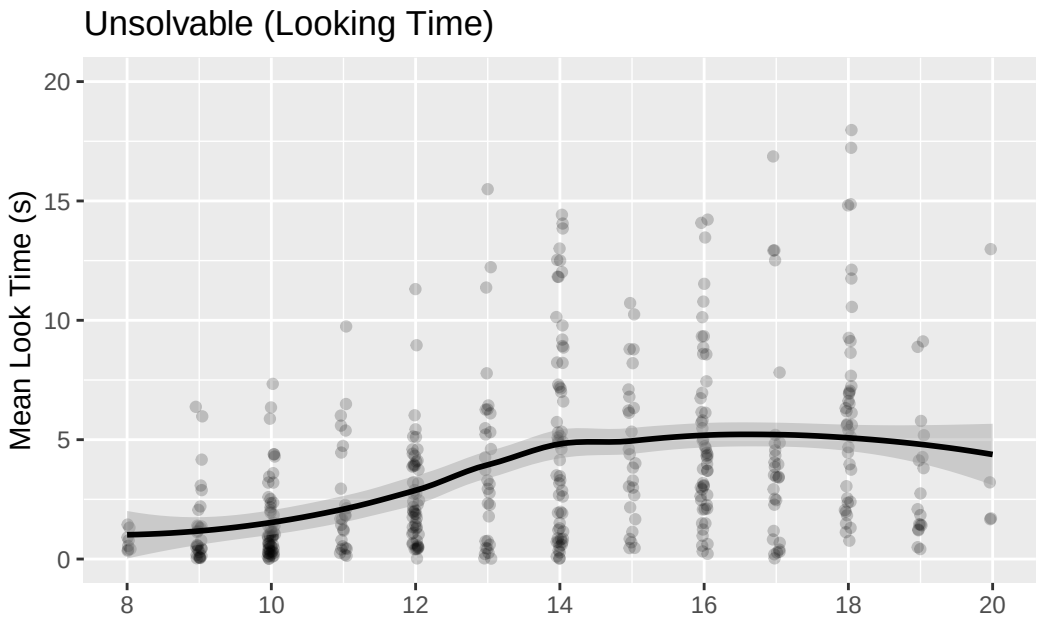



```
print(looktime_plot)
```

```
`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 3 rows containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 51 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
workingmem_long <- workingmem |>
  filter(Status=="Longitudinal") |>
   mutate(
    `Age (wks)` = as.numeric(`Age (wks)`),
    `Proportion Correct` = as.numeric(`Proportion Correct`),
     `experience` = as.numeric(`# Prior Test Sessions`)
  )

workingmem_plot <- workingmem_long|>
  ggplot(aes(x=`Age (wks)`, y = `Proportion Correct`))+
  geom_jitter(alpha=0.2, width = 0.05, height = 0.05)+
  geom_smooth(method="loess", color = "black") +
  scale_x_continuous(
    name ="",
    limits = c(8, 20),
    breaks = seq(8, 20, by = 2)
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, by = 0.25),
    labels = as.character
  ) +
  labs(title = "Working Memory (Delay)")
```

```
print(workingmem_plot)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 146 rows containing missing values or values outside the scale range
(`geom_point()`).
```

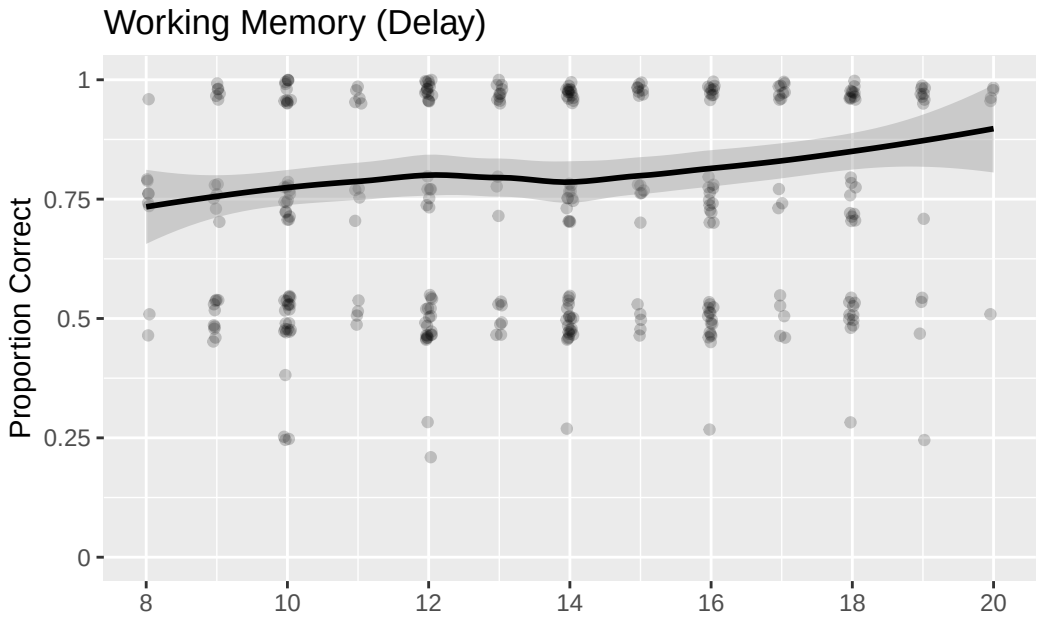


Figure 2

```
(marker_plot   | workingmem_plot | distraction_plot) /
(auditory_plot | odor_plot       | detour_plot)      /
(reversal_plot | looktime_plot   | proplooked_plot)  /
(causal_plot   | momentary_plot  | plot_spacer())&
  theme_grey(base_family = "serif") + #match theme to original
  theme(
```

```
    plot.title = element_text(hjust = 0.5, size = 11),
    axis.line  = element_line(colour = "black"),
    axis.title = element_text(size = 10),
    axis.text  = element_text(size = 9, colour = "black")
  )
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 179 rows containing missing values or values outside the scale range
(`geom_point()`).
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 144 rows containing missing values or values outside the scale range
(`geom_point()`).
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 105 rows containing missing values or values outside the scale range
(`geom_point()`).
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 65 rows containing missing values or values outside the scale range
(`geom_point()`).
```

```
`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 41 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 178 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 83 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 3 rows containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 41 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 125 rows containing missing values or values outside the scale range
(`geom_point()`).
```

```
`geom_smooth()` using formula = 'y ~ x'

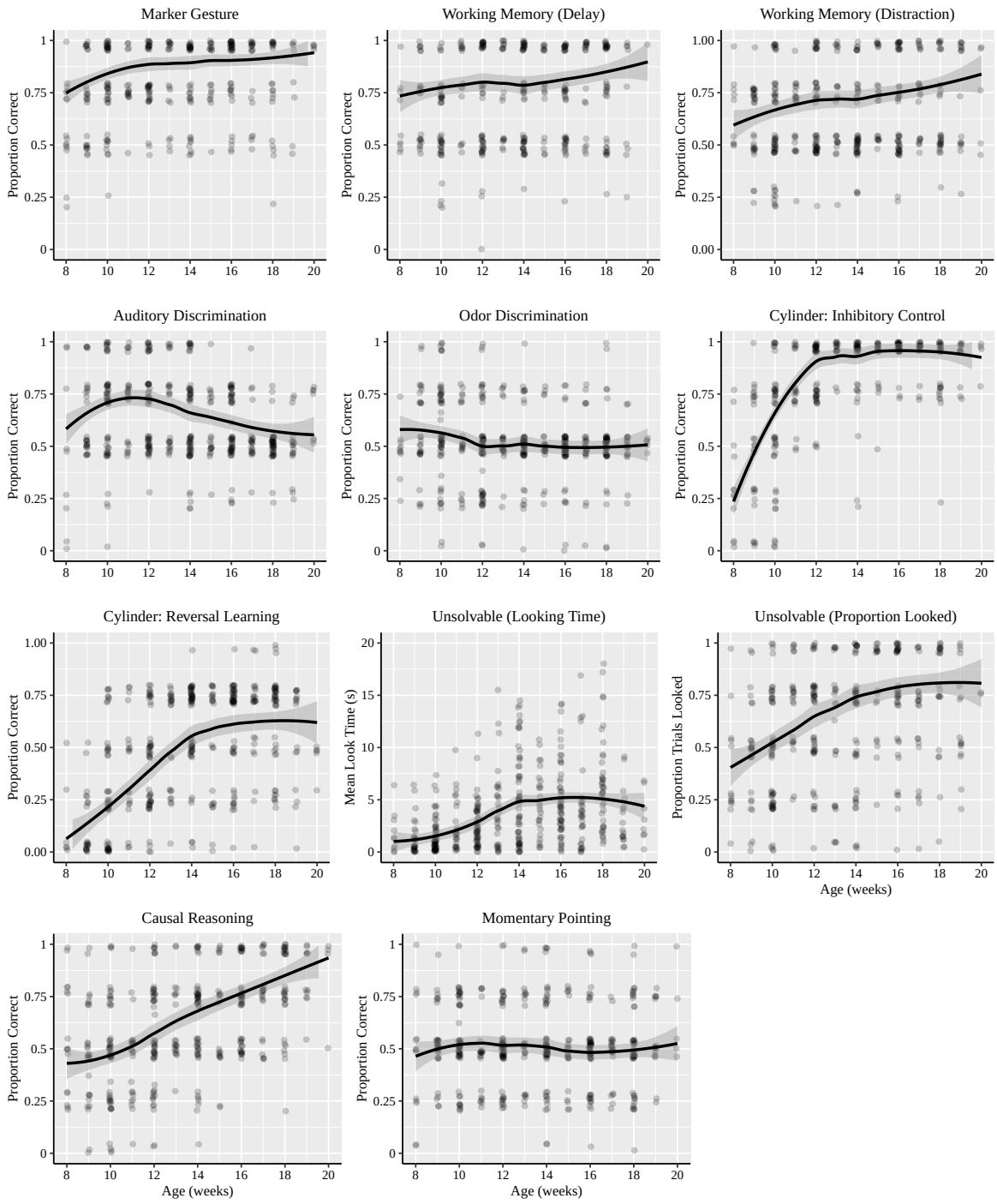
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 90 rows containing missing values or values outside the scale range
(`geom_point()`).

`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 34 rows containing missing values or values outside the scale range
(`geom_point()`).
```


Marker Gesture
Working Memory (Delay)
Working Memory (Distraction)
Auditory Discrimination
Odor Discrimination
Cylinder: Inhibitory Control
Cylinder: Reversal Learning
Unsolvable (Looking Time)
Unsolvable (Proportion Looked)
Causal Reasoning
Momentary Pointing
Proportion Correct
Mean Look Time (s)
Proportion Trials Looked
Age (weeks)

Per-Task Models (Longitudinal)

```
task_models <- function(df) {

m_age <- glmer(cbind(`Trials Correct`, `Total Trials` - `Trials Correct`) ~ `Age (wks)` + (1
                 family = binomial, data = df)
m_exp <- glmer(cbind(`Trials Correct`, `Total Trials` - `Trials Correct`) ~ `Age (wks)` + exp
                 family = binomial, data = df)

tibble(AIC_age = AIC(m_age), AIC_age_exp = AIC(m_exp))
  }

aic_tibble <- bind_rows(
  "Marker gesture"               = task_models(marker_long),
  "Working memory (Delay)"       = task_models(workingmem_long),
  "Working memory (Distraction)" = task_models(distraction_long),
  "Auditory discrimination"      = task_models(auditory_long),
  "Odor discrimination"         = task_models(odor_long),
  "Cylinder: inhibitory control" = task_models(detour_long),
  "Cylinder: reversal learning"  = task_models(reversal_long),
  "Causal reasoning"             = task_models(causal_long),
  "Momentary pointing"           = task_models(momentary_long),
  .id="Task") |>
  mutate(Best = if_else(AIC_age_exp < AIC_age, "age + experience", "age"))
```

```
boundary (singular) fit: see help('isSingular')
boundary (singular) fit: see help('isSingular')
boundary (singular) fit: see help('isSingular')
boundary (singular) fit: see help('isSingular')
```

```
aic_tibble
```

```
# A tibble: 9 x 4
  Task                          AIC_age AIC_age_exp Best
  <chr>                           <dbl>       <dbl> <chr>
1 Marker gesture                   844.        846. age
2 Working memory (Delay)          1050.       1048. age + experience
3 Working memory (Distraction)    1168.       1169. age
4 Auditory discrimination         1226.       1221. age + experience
5 Odor discrimination             1167.       1169. age
```

```
6 Cylinder: inhibitory control    938.        895. age + experience
7 Cylinder: reversal learning    1312.       1298. age + experience
8 Causal reasoning               1210.       1186. age + experience
9 Momentary pointing             1219.       1219. age + experience
```

```
m_age <- glmer(cbind(`# Trials Looked`, `Total Trials` - `# Trials Looked`) ~ `Age (wks)` +
               family = binomial, data = unsolvable_long)
m_exp <- glmer(cbind(`# Trials Looked`, `Total Trials` - `# Trials Looked`) ~ `Age (wks)` + e
               family = binomial, data = unsolvable_long)

aic_tibble2 <- tibble(AIC_age = AIC(m_age), AIC_age_exp = AIC(m_exp))|>
  mutate(Best = if_else(AIC_age_exp < AIC_age, "age + experience", "age"))

aic_tibble2
```

```
# A tibble: 1 x 3
  AIC_age AIC_age_exp Best
    <dbl>       <dbl> <chr>
1   1352.       1353. age
```

```
m_age <- lmer(`Mean Look Time` ~ `Age (wks)` + (1 | `Dog ID`),
              data = unsolvable_long, REML = FALSE)
m_exp <- lmer(`Mean Look Time` ~ `Age (wks)` + experience + (1 | `Dog ID`),
              data = unsolvable_long, REML = FALSE)


aic_tibble3 <- tibble(AIC_age = AIC(m_age), AIC_age_exp = AIC(m_exp))|>
  mutate(Best = if_else(AIC_age_exp < AIC_age, "age + experience", "age"))

aic_tibble3
```

```
# A tibble: 1 x 3
  AIC_age AIC_age_exp Best
    <dbl>       <dbl> <chr>
1   2514.       2515. age
```

`correct ~ age + (1 | puppy_ID)` is better for:

- Marker Gesture (AIC = 844.3)
- Working Memory with Distraction (AIC = 1168.1)

- Odor Discrimination (AIC = 1166.9)
- Unsolvable Proportion Looked (AIC = 1351.6)
- Unsolvable Average Looking Time (AIC = 2514.5).

`correct ~ age + experience + (1 | puppy_ID)` is better for:

- Working Memory with Delay (AIC = 1047.9)
- Auditory Discrimination (AIC = 1220.9)
- Cylinder Inhibitory Control (AIC = 894.9)
- Cylinder Reversal Learning (AIC = 1297.8)
- Causal Reasoning (AIC = 1185.8).
- Momentary Pointing (AIC = 1218.8).

Per-Task Models (Longitudinal + Control)

```
task_models2 <- function(df2) {

m_age <- glmer(cbind(`Trials Correct`, `Total Trials` - `Trials Correct`) ~ `Age (wks)` + (1
                 family = binomial, data = df2)
m_exp <- glmer(cbind(`Trials Correct`, `Total Trials` - `Trials Correct`) ~ `Age (wks)` + `#
                 family = binomial, data = df2)

tibble(AIC_age = AIC(m_age), AIC_age_exp = AIC(m_exp))
  }

aic_tibble4 <- bind_rows(
  "Marker gesture"                = task_models2(marker),
  "Working memory (Delay)"        = task_models2(workingmem),
  "Working memory (Distraction)" = task_models2(distraction),
  "Auditory discrimination"       = task_models2(auditory),
  "Odor discrimination"          = task_models2(odor),
  "Cylinder: inhibitory control" = task_models2(detour),
  "Cylinder: reversal learning"  = task_models2(reversal),
  "Causal reasoning"              = task_models2(causal),
  "Momentary pointing"            = task_models2(momentary),
  .id="Task") |>
  mutate(Best = if_else(AIC_age_exp < AIC_age, "age + experience", "age"))
```

```
boundary (singular) fit: see help('isSingular')
boundary (singular) fit: see help('isSingular')
boundary (singular) fit: see help('isSingular')
boundary (singular) fit: see help('isSingular')
```

```
aic_tibble4
```

```
# A tibble: 9 x 4
  Task                          AIC_age AIC_age_exp Best
  <chr>                           <dbl>       <dbl> <chr>
1 Marker gesture                   876.        878. age
2 Working memory (Delay)          1081.       1073. age + experience
3 Working memory (Distraction)    1223.       1224. age
4 Auditory discrimination         1285.       1277. age + experience
5 Odor discrimination             1226.       1227. age
6 Cylinder: inhibitory control    1089.        957. age + experience
7 Cylinder: reversal learning     1427.       1364. age + experience
8 Causal reasoning                1327.       1244. age + experience
9 Momentary pointing              1269.       1270. age
```

```
m_age <- glmer(cbind(`# Trials Looked`, `Total Trials` - `# Trials Looked`) ~ `Age (wks)` +
               family = binomial, data = unsolvable)
m_exp <- glmer(cbind(`# Trials Looked`, `Total Trials` - `# Trials Looked`) ~ `Age (wks)` +
               family = binomial, data = unsolvable)

aic_tibble5 <- tibble(AIC_age = AIC(m_age), AIC_age_exp = AIC(m_exp))|>
  mutate(Best = if_else(AIC_age_exp < AIC_age, "age + experience", "age"))

aic_tibble5
```

```
# A tibble: 1 x 3
  AIC_age AIC_age_exp Best
    <dbl>       <dbl> <chr>
1   1443.       1431. age + experience
```

```
m_age <- lmer(`Mean Look Time` ~ `Age (wks)` + (1 | `Dog ID`),
              data = unsolvable, REML = FALSE)
m_exp <- lmer(`Mean Look Time` ~ `Age (wks)` + `# Prior Test Sessions` + (1 | `Dog ID`),
              data = unsolvable, REML = FALSE)
```

```r
aic_tibble6 <- tibble(AIC_age = AIC(m_age), AIC_age_exp = AIC(m_exp))|>
  mutate(Best = if_else(AIC_age_exp < AIC_age, "age + experience", "age"))

aic_tibble6
```

```
# A tibble: 1 x 3
  AIC_age AIC_age_exp Best
    <dbl>       <dbl> <chr>
1   2640.       2629. age + experience
```

`correct ~ age + (1 | puppy_ID)` is better for:

- Marker Gesture (AIC = 876.1)
- Working Memory with Distraction (AIC = 1223.1)
- Odor Discrimination (AIC = 1225.8)
- Momentary Pointing (AIC =)

`correct ~ age + experience + (1 | puppy_ID)` is better for:

- Working Memory with Delay (AIC = 1072.8)
- Auditory Discrimination (AIC = 1276.8)
- Cylinder Inhibitory Control (AIC = 957.2)
- Cylinder Reversal Learning (AIC = 1364.4)
- Causal Reasoning (AIC = 1244.3).
- Unsolvable Proportion Looked (AIC = 1431.1)
- Unsolvable Average Looking Time (AIC = 2629.2).